HK IMO Training 2021-2022
Problem Set 11 Suggested Solutions

Problem 41 (Saudi Arabia IMO TST 2016 Level 4 Day 1 Problem 1). Let $n \geqslant 3$ be an integer. Let $a_1, a_2, \ldots, a_n$ be n pairwise distinct integers. Find the smallest possible value of

$$(a_1 - a_2)^2 + (a_2 - a_3)^2 + \cdots + (a_{n-1} - a_n)^2 + (a_n - a_1)^2.$$

Solution. The smallest possible value is $4n - 6$.

Firstly, we prove by induction that

$$(a_1 - a_2)^2 + (a_2 - a_3)^2 + \cdots + (a_{n-1} - a_n)^2 + (a_n - a_1)^2 \geqslant 4n - 6. \tag{1}$$

When $n = 3$, each of $|a_1 - a_2|$, $|a_2 - a_3|$ and $|a_3 - a_1|$ is at least 1, and one of them is at least 2. Therefore, the sum of squares is at least $1^2 + 1^2 + 2^2 = 6 = 4(3) - 6$. This proves the base case.

Assume (1) holds for some $n = k \geqslant 3$. When $n = k + 1$, WLOG assume a_{k+1} is the largest among $a_1, a_2, \ldots, a_{k+1}$, and assume $a_k > a_1$. Then we have

$$\begin{aligned}
&(a_1 - a_2)^2 + (a_2 - a_3)^2 + \cdots + (a_k - a_{k+1})^2 + (a_{k+1} - a_1)^2 \\
&= [(a_1 - a_2)^2 + (a_2 - a_3)^2 + \cdots + (a_{k-1} - a_k)^2 + (a_k - a_1)^2] + (a_k - a_{k+1})^2 + (a_{k+1} - a_1)^2 - (a_k - a_1)^2 \\
&\geqslant 4k - 6 + (a_{k+1} - a_k)^2 + (a_{k+1} - a_1)^2 - (a_k - a_1)^2 \\
&\geqslant 4k - 6 + 1^2 + (a_k + 1 - a_1)^2 - (a_k - a_1)^2 \\
&= 4k - 4 + 2(a_k - a_1) \\
&\geqslant 4k - 2.
\end{aligned}$$

This proves the inductive step. Thus, (1) is proved by induction.

Secondly, we provide an equality case. For odd n, we can take $a_1, a_2, \ldots, a_n = 1, 3, 5, \ldots, n, n-1, n-3, \ldots, 2$. For even n, we can take $a_1, a_2, \ldots, a_n = 1, 3, 5, \ldots, n-1, n, n-2, \ldots, 2$. In either case, we have

$$(a_1 - a_2)^2 + (a_2 - a_3)^2 + \cdots + (a_{n-1} - a_n)^2 + (a_n - a_1)^2 = 2 \cdot 1^2 + (n-2) \cdot 2^2 = 4n - 6.$$

□

Problem 42 (Benelux MO 2014 Problem 2). Let n be a positive integer. There are $4n$ coins lying in a row, including $2n$ one-dollar coins and $2n$ two-dollar coins. In each move, we can choose k consecutive one-dollar coins and choose k consecutive two-dollar coins, and swap these two groups of coins, where k can be any positive integer that can vary in different moves. Find the minimum number of moves needed so that we can always reach the state in which the first $2n$ coins are one-dollar coins, regardless of the initial positions of the coins.

Solution. The minimum number of moves needed is n.

Firstly, we prove that n moves always suffice. Suppose there are m one-dollar coins among the first $2n$ coins initially. If $m \geqslant n$, then there are $2n - m \leqslant n$ two-dollar coins among the first $2n$ coins. We can swap each of these two-dollar coins with a one-dollar coin among the last $2n$ coins one by one. Afterwards, the first $2n$ coins are all one-dollar coins, and at most n moves are used.

If $m \leqslant n - 1$, we can swap each of these one-dollar coins with a two-dollar coin among the last $2n$ coins one by one. Afterwards, the first $2n$ coins are all two-dollar coins, and at most $n - 1$ moves are used. Using one more move, we can swap all the one-dollar coins and all the two-dollar coins, yielding the desired configuration. Therefore, at most n moves are needed.

Secondly, consider the initial configuration $1, 2, 1, 2, \ldots, 1, 2$. There are $4n - 1$ pairs of consecutive coins of distinct values initially. In each move, we can reduce this number by at most 4 since we only swap two groups of coins in each move, and only the pairs at the two ends of a group may be affected. In the end, there is only 1 pair of consecutive coins of distinct values. Therefore, at least $\dfrac{(4n-1)-1}{4} = n - \dfrac{1}{2}$ moves are needed. As the number of moves is an integer, at least n moves are needed. This proves the result.

□

Problem 43 (Saudi Arabia IMO TST 2018 Day 3 Problem 1)**.** Let $a_n = 2^{n^2+1} - 3^n$ for every positive integer n.

(a) Prove that there are infinitely many prime numbers dividing some of the a_n's.

(b) Prove that there are infinitely many prime numbers not dividing any of the a_n's.

Solution.

(a) Suppose on the contrary that only finitely many primes $p_1, p_2, \ldots, p_k$ may divide some of the a_n's. Note that $p_j \neq 3$ for any j. Also, as $a_2 = 23$, the prime 23 is one of them. Consider $n = (p_1 - 1)(p_2 - 1)\cdots(p_k - 1) > 1$. Then we have $a_n > 1$ but

$$a_n = 2^{n^2+1} - 3^n \equiv 2^1 - 1 = 1 \not\equiv 0 \pmod{p_j}$$

for every j by the Euler-Fermat theorem. This contradicts the assumption that all prime divisors of a_n are p_j. So there are infinitely many prime numbers dividing some of the a_n's.

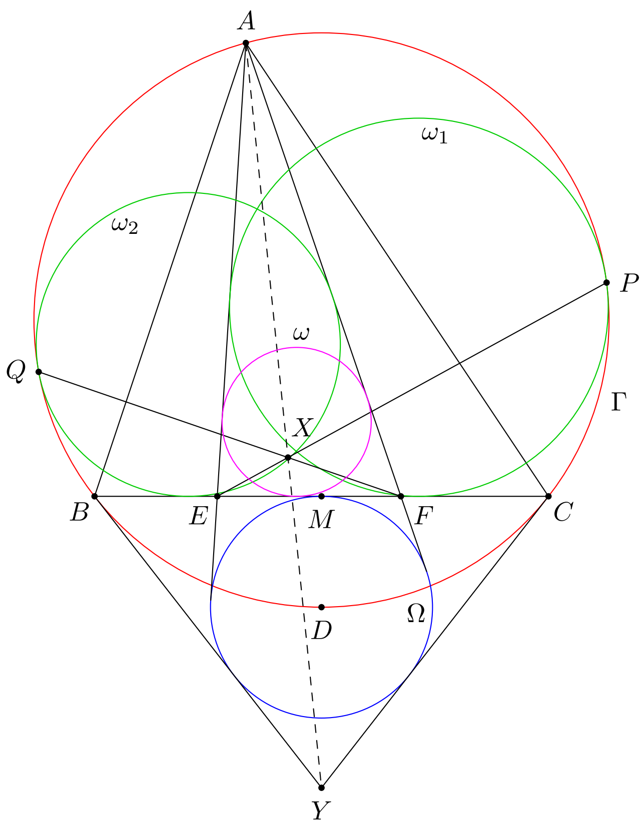
(b) By Dirichlet's theorem, there are infinitely many primes $p \equiv 5 \pmod{24}$. We claim that all such primes p do not divide any of the a_n's. Suppose on the contrary that $2^{n^2+1} \equiv 3^n \pmod{p}$ for some n.

If n is even, then 2^{n^2} and 3^n are quadratic residues modulo p. Thus, $2 \equiv 3^n 2^{-n^2} \pmod{p}$ is a quadratic residue modulo p. But this is not true since $p \equiv -3 \pmod{8}$.

If n is odd, then 2^{n^2+1} and 3^{n-1} are quadratic residues modulo p. Thus, $3 \equiv 2^{n^2+1}3^{-(n-1)} \pmod{p}$ is a quadratic residue modulo p. But this is not true since $\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)(-1)^{\frac{(p-1)(3-1)}{4}} = \left(\frac{p}{3}\right) = \left(\frac{2}{3}\right) = -1$.

□

Problem 44 (RMM Shortlist 2019 G4)**.** Let Γ be the circumcircle of an acute $\triangle ABC$. Let M be the midpoint of BC, and let D be the midpoint of the minor arc BC of Γ. Let Ω be the circle with centre D that passes through M. The two tangents from A to Ω meet the side BC at E and F respectively, with E lying between B and F. A circle ω_1 is tangent to the segments AE and CE, and is also tangent to Γ at P. A circle ω_2 is tangent to the segments AF and BF, and is also tangent to Γ at Q. Let X be the intersection point of EP and FQ. Prove that $\angle EAX = \angle MAF$.



Solution. Let Y be the intersection point of the tangents at B and C to Γ. Since $\angle YBD = \angle BCD = \angle CBD$, BD is the internal angle bisector of $\angle YBC$. Similarly, CD is the internal angle bisector of $\angle YCB$. Thus, D is the incentre of $\triangle YBC$. So YB and YC are tangent to Ω.

Let ω be the incircle of $\triangle AEF$, and let X' be the external centre of similitude of ω and Γ. Applying Monge's theorem to ω, ω_2 and Γ, the points F, Q, X' are collinear. Similarly, applying Monge's theorem to ω, ω_1 and Γ, the points E, P, X' are collinear. Therefore, X' is the intersection point of FQ and EP, which means $X' = X$.

Lastly, applying Monge's theorem to ω, Ω and Γ, the points A, X, Y are collinear. It is well-known that AY is the A-symmedian of $\triangle ABC$. So AX and AM are isogonals w.r.t. $\angle BAC$. Also, since D is the A-excentre of $\triangle AEF$, AD is the internal angle bisector of $\angle EAF$. But then AD is the internal angle bisector of $\angle BAC$. Therefore, AE and AF are isogonals w.r.t. $\angle BAC$. It follows that $\angle EAX = \angle MAF$ as desired. $\square$